Based on the provided JavaScript source material, here are several "predict-the-output" questions and answers suitable for exam preparation.

Question 1: Variable Scopes with var

What is the output of the following code?

```
var i = "function scope";
if (true) {
  var i = "block scope";
  console.log("Inside:", i);
}
console.log("Outside:", i);
```

Answer: Inside: block scopeOutside: block scope**Explanation:** When you declare a variable with var, it is declared globally or locally if inside a function. Because it is not block-scoped, the assignment inside the if statement overwrites the variable in the outer scope 1-3.

Question 2: String Immutability

What is the output of the following code?

```
let myStr = "Jello World";
myStr = "H";
console.log(myStr);
```

Answer: Jello World**Explanation:** Strings in JavaScript are immutable, meaning they cannot be altered once created. While the variable itself can be reassigned to a new string, individual characters of a string literal cannot be changed via bracket notation 4, 5.

Question 3: Equality vs. Strict Equality

What is the output of the following code?

```
function compare(a, b) {
  if (a === b) {
    return "Strict Equal";
  } else if (a == b) {
    return "Equal";
  } else {
    return "Not Equal";
  }
}
console.log(compare(10, "10"));
```

Answer: Equal**Explanation:** The equality operator (==) performs a type conversion, treating the number 10 and the string "10" as the same. The strict equality operator (===) does not perform type conversion and requires both the value and the type to be identical 6-9.

Question 4: Array Manipulation

What is the output of the following code?

```
let arr = [10-12];
arr.push(4);
let removed = arr.shift();
console.log(arr);
console.log(removed);
```

Answer:11-131**Explanation:** The .push() function appends data to the end of an array 14. The .shift() function removes the first element of the array and returns it 15.

Question 5: Local vs. Global Scope

What is the output of the following code?

```
var outerWear = "T-Shirt";
function myOutfit() {
  var outerWear = "sweater";
  return outerWear;
}
console.log(myOutfit());
console.log(outerWear);
```

Answer:sweaterT-Shirt**Explanation:** It is possible to have both local and global variables with the same name. When this occurs, the local variable inside the function takes precedence over the global variable 16-18.

Question 6: The Rest Operator

What is the output of the following code?

```
const countArgs = (...args) => {
  return args.length;
};
console.log(countArgs(1, 2, 3, 4, 5));
```

Answer:5**Explanation:** The rest operator (...) allows a function to accept a variable number of arguments and converts them into a single array named args 19, 20.

Question 7: Destructuring Arrays

What is the output of the following code?

```
const [x, y, , z] = [10-13, 21, 22];
console.log(x, y, z);
```

Answer:1 2 4**Explanation:** Destructuring assignment from arrays assigns variables in order. By using commas with nothing in between, you can skip specific elements in the array 23, 24.

Question 8: Object Property Access

What is the output of the following code?

```
const testObj = {
  "an entree": "hamburger",
  "the drink": "water"
```

```
};
const drinkValue = testObj["the drink"];
console.log(drinkValue);
```

Answer:water**Explanation:** While dot notation is common for accessing object properties, bracket notation is required if the property name has a space in it 25, 26.

Question 9: Ternary Operators

What is the output of the following code?

```
function checkSign(num) {
  return (num > 0) ? "positive" : (num < 0) ? "negative" : "zero";
}
console.log(checkSign(0));
```

Answer:zero**Explanation:** This code uses nested conditional (ternary) operators. Since 0 > 0 is false, it moves to the second condition. Since 0 < 0 is also false, it returns the final fallback value, "zero" 27-29.

Question 10: Object.freeze

What is the output of the following code?

```
const MATH_CONSTANTS = { PI: 3.14 };
Object.freeze(MATH_CONSTANTS);
try {
  MATH_CONSTANTS.PI = 99;
} catch(ex) {
  console.log("Error caught");
}
console.log(MATH_CONSTANTS.PI);
```

Answer: Error caught3.14**Explanation:** Declaring an object with const does not prevent mutation. Object.freeze is used to lock an object so that its properties cannot be added, deleted, or changed 30-32.